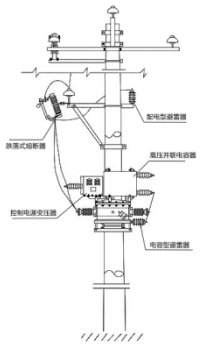
Product Manual for Smart Cutout

Product Introduction

Smart Cutout

Product

Smart Cutout switches are the most commonly used short-circuit protection switches for 10kV distribution line branch circuits and distribution transformers. As core components for overload and short-circuit protection, their working principle is based on a dual mechanism of 'thermal effect of the Cutout' and 'mechanical dropping'. When the current exceeds the rated threshold, the silver-copper alloy Cutout inside melts due to Joule heating, triggering a spring mechanism that causes the Cutout tube to automatically drop under



Technical principles and system

System Architecture	1. A Cutout body module, equipped with an internal current sensor and mechanical drive mechanism; 2. The wireless communication module supports multi-mode communication including LoRa, 4G, and 5G.
Technical principles	When line overload or short circuit is detected, the control unit receives dispatch commands via a wireless channel, triggering the high-speed permanent magnet mechanism to complete the disconnection operation, with an action time controllable within 20 milliseconds. Using frequency-hopping spread spectrum (FHSS) technology ensures the interference resistance of command transmission, with a maximum communication distance meters.

Key technologic

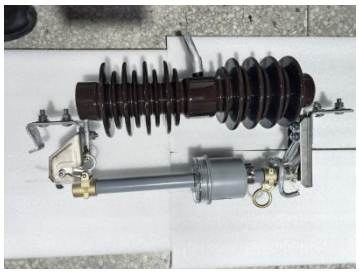
- Bidirectional communication verification mechanism: uses AES-256 encryption protocol combined with a three-way handshake confirmation process to avoid false actions.
- Adaptive protection algorithm: trains AI models based on historical fault data to dynamically adjust inverse-time characteristics.
- Energy self-sustaining design: utilizes power from current transformers combined with supercapacitors for energy storage, ensuring continuous operation for 72 hours without external

Benefit

- Operational efficiency has been improved: pole climbing operations have decreased by 80%, and the time for each operation has sharply reduced from 45 minutes to 10 seconds.
- Safety risk control: incidents of arc burns caused by live-line operations have been eliminated.
- Fault location accuracy: combined with the SCADA system, fault section location accuracy can be enhanced to ±15 meters.

XNPD-T803DJ Smart Cutout

Product



The XNPD-T803DJ Smart Cutout, suitable for 10kV and 20kV overhead line user boundary points. The product fully leverages new technologies such as the power Internet of Things (IoT), AC sensors, and deep integration of primary and secondary systems to innovatively integrate high-voltage Cutout protection, position signal detection, load current monitoring, and fault detection into a single device. It enables real-time collection and remote transmission of operational status data from the high-voltage side of overhead distribution substations, and supports abnormal condition and fault alarms. The product features easy installation, no need for power outages during installation, and simple maintenance.

The XNPD-T803DJ Smart Cutout uses current sampling to collect high-voltage side Cutout position data, current values, and phase-to-phase short-circuit alarms. It communicates with communication management devices via a low-power wireless network, transmitting data to the distribution automation master station system through the communication management device or

Technical

（1） Direct replacement of existing cutout tubes to achieve remote control and remote signalling automation;

（2） Low current power consumption and low power design: 0.4A startup current, 0.3A maintenance current, and equipped with a supercapacitor for power storage, ensuring the battery serves as a true backup power source and significantly extending product lifespan;

(3) AC current sensor (CT) with open-circuit protection for enhanced safety, measurement accuracy of 0.5 grade;

（4） Wireless maintenance and wireless firmware upgrades;

Main

（1） Collect the on/off position signals of the high-voltage cutout elements

（2） Collect three-phase current

（3） Phase-to-phase short-circuit alarm

（4） Remote communication functionality, with communication protocols compliant with IEC 60870-5-101 and IEC 60870-5-104

XNPD-T803DJ

Smart Cutout

Technical

Name			Standard parameter value	
Product Model			XNPD-T803DJ	
Operating Environment	Operating Temperature		-40℃ to 70℃	
	Humidity		10% to 100%	
High-voltage Cutout				
Rated voltage	12kV		24kV	
Rated current (A)	10	200	100	200
Rated breaking current (A)	10000	12500	8000	9600
Rated withstand voltage (kV)	35		45	
Lightning impulse withstand voltage (kV)	110		125	
Creepage distance (mm)	230		320	
Smart Cutout monitoring unit				
Operating power supply	Main power supply		Current self-supply	
	Backup Power Supply		Disposable lithium battery 3.6V, 4Ah	
			Supercapacitor	
Power consumption	Static		≤80 μA	
Measurement and control capacity			(1) Telemetry: Phase current (2) Telecontrol: Open/closed position signal	
Current measurement accuracy			Phase measurement value: Class 0.5 Phase protection value ≤3%	
Protection rating			IP65	
Smart Cutout Communication Management Device				
Operating Power Supply	Main power supply		AC 220V	
	Backup Power Supply		Supercapacitor	
Power Consumption	Static		≤ 3VA	
Communication interface	Maintenance RS232		9600 bit/s	
	Communication RS232		9600 bit/s	
	RS485		9600 bit/s	
Remote communication	Communication protocol		Supports 4G/3G/2G adaptive	
	Communication protocol		IEC 60870-5-101 IEC 60870-5-104	
Protection Rating	Local communication type		IP3X	
	Remote communication		IP55	

Product Introduction

XNPD-T803DQ integrated smart Cutout

Product

The XNPD-T803DQ integrated smart Cutout is designed for branch and boundary points of 10kV overhead power lines. The product fully leverages new technologies such as power Internet of Things (IoT), AC sensors, and deep integration of primary and secondary systems to innovatively integrate functions such as high-voltage Cutout position signal detection, high-voltage side current, voltage, power, and power factor measurement, phase-to-phase short-circuit alarms, and single-phase ground fault alarms. It enables real-time monitoring and remote transmission of operational status of high-voltage sides in 10kV overhead line distribution substations, supports abnormal and fault alarms, and features convenient installation and simple maintenance.

The XNPD-T803DQ integrated smart Cutout integrates a current sensor, voltage sensor, and capacitive voltage divider into a single casting, enabling the collection of high-voltage side Cutout position signals, three-phase and zero-sequence current signals, three-phase and zero-sequence voltage signals, active and reactive power factor signals, line loss signals, phase-to-phase short-circuit alarms,

Technical

（1）Deep integration of primary and secondary circuits, with an integrated cast-moulded current sensor, voltage sensor, and capacitive voltage divider power extraction device

（2）Standardised Cutout holders for higher reliability and easier maintenance

（3）Capacitive voltage division self-power supply, safe and reliable, suitable for any application scenario.

（4）Independent measurement and control unit, allowing online hot-swappable installation, simplifying maintenance;

Main

（1）Collects the on/off position signals of high-voltage cutout load-breaking components

（2）Collect three-phase and zero-sequence current, three-phase and zero-sequence voltage, active power, reactive power, power factor, frequency, etc.

（3）Phase-to-phase short-circuit alarm, single-phase ground fault alarm

（4）Remote communication functionality, supporting communication protocols including IEC 60870-5-101 and IEC 60870-5-104.

XNPD-T803DQ
integrated smart Cutout

Technical

Name		Unit	Standard Parameter Value
High-voltage intelligent disconnect switch device (primary part)			
Rated Voltage		kV	1
Rated current		A	100
Rated short-circuit breaking current		kA	12.5
Rated lightning impulse withstand voltage (peak)	Earth/phase	kV	75
	Between breakers (without load-breaking components)	kV	85
Rated 1-minute power frequency withstand voltage	Ground/phase-to-phase	kV	Dry flash 42 kV
		kV	Wet flash 30 kV
	Break point (no load Cutout)	kV	Dry flash 49 kV
		kV	Wet flash 36 kV
External insulation			Outdoor epoxy resin insulation
Cutout type			K
Melting component exhaust method			Single-end downward exhaust
Line high-voltage capacitor power supply			
Rated voltage		kV	10/√3
Rated output voltage		V	1
Starting voltage (phase voltage)		kV	≤80% of Un
Rated output power		VA	≥2
Frequency		Hz	50
Power frequency withstand voltage (whole machine test)			42 kV/1 min

XNPD-T803DQ
integrated smart Cutout

Technical

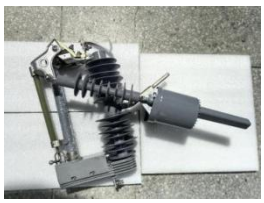
High-voltage intelligent disconnect switch device controller (secondary part)			
Electronic Analog Input Nominal Value	Phase Voltage	V	3.25/√3
	Zero-sequence voltage	V	6.5/3
	Phase current	V	1
	Zero-sequence current	V	0.2
Operating power supply		V	AC 15 V, three-phase
Switching control capacity			Remote measurement: Collects 3 phase voltages, 3 phase currents, zero-sequence current, and zero-sequence voltage Remote signalling: Three-phase position signals;
Voltage measurement accuracy (key indicator)		Grade	Phase voltage: ≤0.5% (Class 0.5) Zero-sequence voltage: ≤0.5% (Class 0.5)
Current measurement accuracy (critical parameter)		Grade	Phase measurement value: 0.5 class (≤1.2In), Phase protection value ≤3% (≤10In), Zero-sequence current 0.5 grade
Active power, reactive power accuracy			≤ 0.5% (Class 0.5)
Time-keeping accuracy			The error should not exceed 2 seconds every 24 hours
Remote communication	Communication protocol		Supports 4G/3G/2G adaptive
	Communication Protocol		IEC 60870-5-101 IEC 60870-5-104
Backup Power Supply Method			Supercapacitor: Maintains the controller and communication module for at least 10 minutes.
Line fault detection			Phase-to-phase short circuit monitoring and alarm Single-phase ground fault monitoring and alarm
Ground fault determination			In systems with an ungrounded neutral point or a neutral point grounded through an arc suppression coil, accurately detect a transition resistance of no less than 4000 ohms.

Product Introduction

XNPD-T803DK Smart Cutout

Product

The XNPD-T803DK Smart Cutout consists of a high-voltage cutout, an electric operating mechanism, and a controller, enabling the collection of position signals for the opening and closing of high-voltage cutouts, as well as remote control and local control functions.
Signals are transmitted to the distribution automation master station system via the controller or a smart grid gateway.
It supports expandable current measurement, enabling three-phase and zero-sequence current measurement, phase-to-phase short-circuit alarms, and single-phase ground fault alarms.



Technical

(1) Does not alter the maintenance and operation methods of the cutout
(2) Supports multiple power supply options, including AC 220V and solar power
(3) Supports multiple communication methods: supports connection to the distribution automation master station system via RS485 interface and 4G communication;

Main

(1) Collects the open/closed position signals of the Cutout elements in high-voltage cutouts
(2) Supports local opening and closing operations, as well as remote control opening and closing operations;
(3) Supports expandable current measurement, enabling three-phase and zero-sequence current measurement, phase-to-phase short-circuit alarms, and single-phase ground fault alarms
(4) Supports remote communication functionality, with communication protocols compliant with IEC 60870-5-101 and IEC 60870-5-104.

XNPD-T803DK

smart Cutout

Technical

Name		Standard Parameter Value	
Product Model		XNPD-T803DK	
Operating Environment	Operating Temperature	-40°C to 70°C	
	Humidity	10% to 100%	
High-voltage cutout			
Rated voltage	12kV	24kV	
Rated current (A)	10 / 200	100	200
Rated breaking current (A)	10000 / 12500	8000	9600
Rated withstand voltage (kV)	35	45	
Lightning impulse withstand voltage (kV)	110	125	
Creepage distance (mm)	230	320	
Control parameters			
Remote control on/off switching cycles		≥ 500 times	
Drive motor input voltage		DC 12V, DC 24V	
Drive motor power		≥120W	
Drive motor instantaneous operating time		Opening 0.5s, closing 0.2s	
Opening action time		2 seconds	
Phase action time interval		2 seconds	
Closing action time		2s	
Smart Cutout controller			
Operating power supply	Main power supply	AC 220V, DC 18V solar	
	Backup Power Supply	Maintenance-free valve-regulated gel battery, 12V/8AH	
Measurement and control capacity		Remote signal: Cutout position signal Remote control: Three-channel output	
Expanded functions: Current measurement accuracy		Class 0.5, 5P10	
Communication interface		RS232	
		RS485	
Remote communication	Communication Protocol	Supports 4G/3G/2G adaptive	
	Communication protocol	IEC 60870-5-101 IEC 60870-5-104	